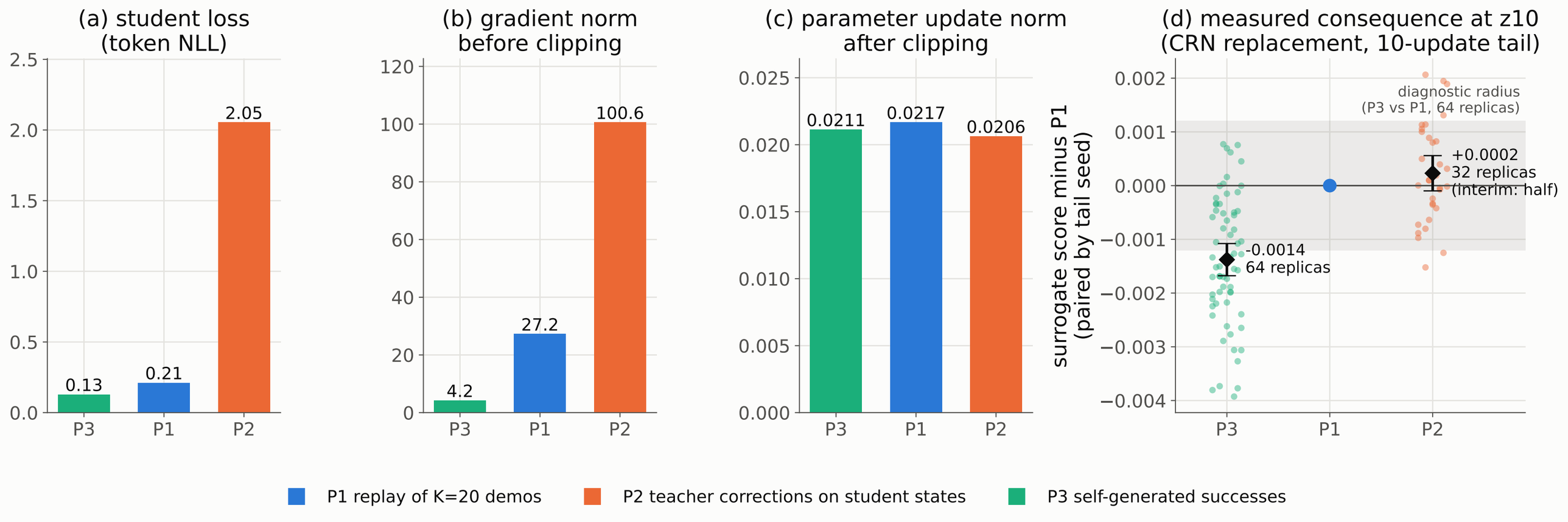
(a) student loss
(token NLL)
2.5
2.0
1.5
1.0
0.5
0.0
0.13
0.21
2.05
P3
P1
P2
(b) gradient norm
before clipping
120
100
80
60
40
20
0
4.2
27.2
100.6
P3
P1
P2
(c) parameter update norm
after clipping
0.025
0.020
0.015
0.010
0.005
0.000
0.0211
0.0217
0.0206
P3
P1
P2
(d) measured consequence at z10
(CRN replacement, 10-update tail)
surrogate score minus P1
(paired by tail seed)
0.002
0.001
0.000
−0.001
−0.002
−0.003
−0.004
diagnostic radius
(P3 vs P1, 64 replicas)
+0.0002
32 replicas
(interim: half)
-0.0014
64 replicas
P3
P1
P2
P1 replay of K=20 demos
P2 teacher corrections on student states
P3 self-generated successes